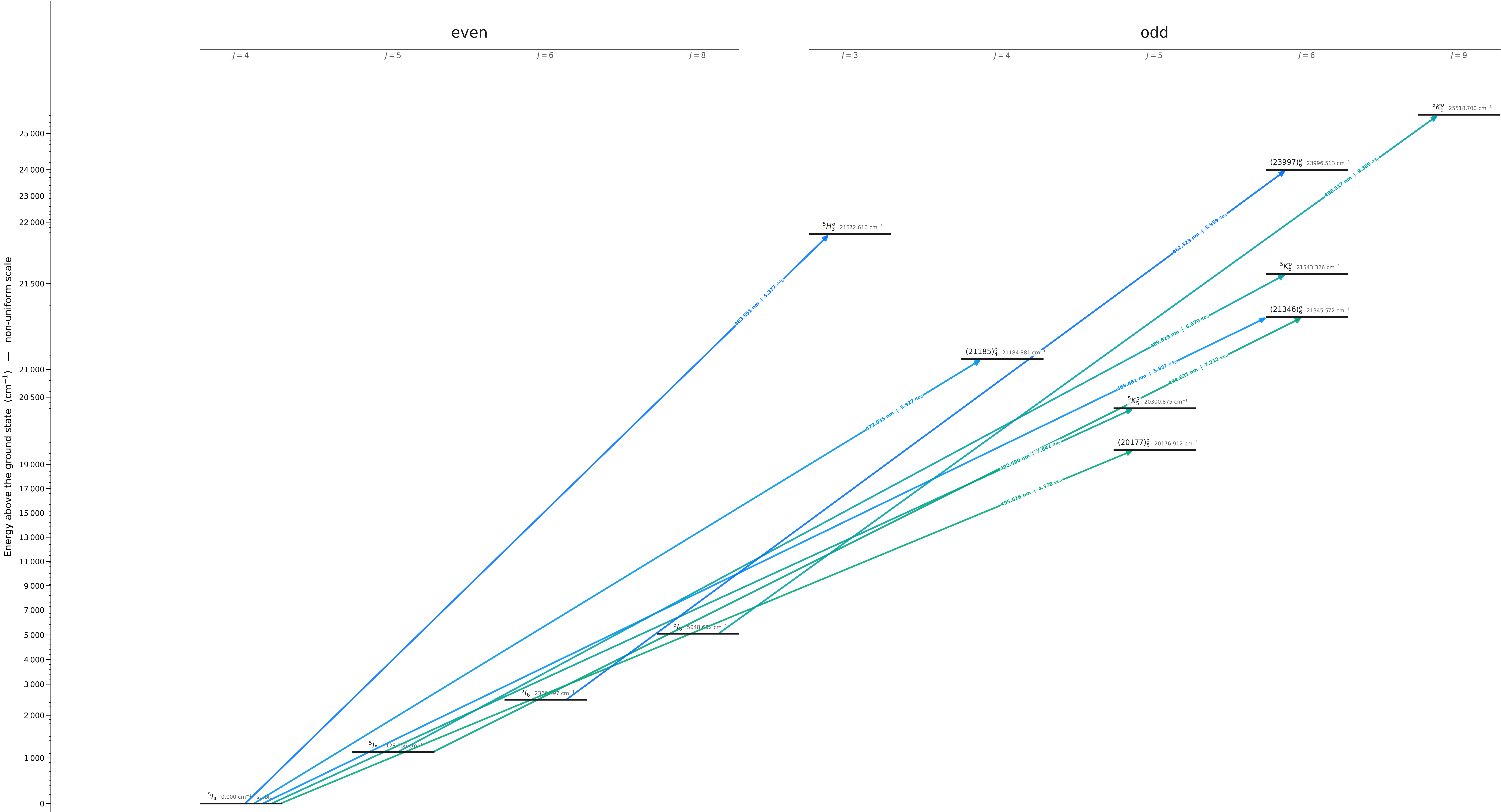
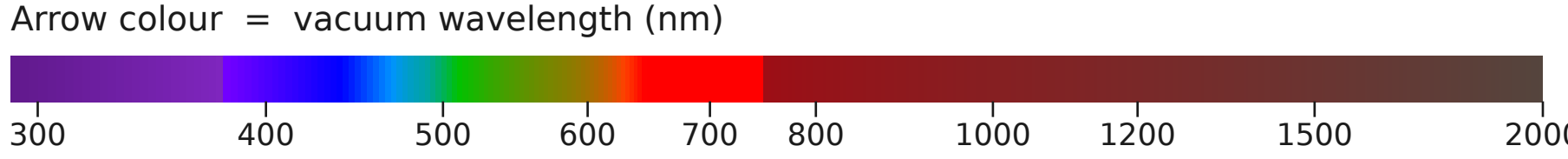


Neodymium

Nd (Nd I). Strongest classified transitions below 2 μm (NIST): 12 levels, 9 lines.

Nd⁺ ionisation limit 44560.11 cm⁻¹ = 5.52475 eV (224.416 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|⟨J||e r||J'⟩|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J e r J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
⁵ I ₆ - (23997) ₆ ^o	462.3226	462.1932	648.4486	5.959	5.600e+07	-	-	-
⁵ I ₄ - ⁵ H ₃ ^o	463.5508	463.4210	646.7306	5.377	8.400e+07	-	-	-
⁵ I ₄ - (21346) ₆ ^o	468.4812	468.3501	639.9242	5.857	5.200e+07	-	-	-
⁵ I ₄ - (21185) ₂ ^o	472.0347	471.9027	635.1068	3.927	3.300e+07	-	-	-
⁵ I ₈ - ⁵ K ₅ ^o	488.5174	488.3810	613.6781	9.809	8.800e+07	-	-	-
⁵ I ₅ - ⁵ K ₆ ^o	489.8294	489.6927	612.0344	6.670	5.900e+07	-	-	-
⁵ I ₄ - ⁵ K ₅ ^o	492.5896	492.4521	608.6049	7.642	9.000e+07	-	-	-
⁵ I ₅ - (21346) ₆ ^o	494.6206	494.4826	606.1059	7.212	6.700e+07	-	-	-
⁵ I ₄ - (20177) ₅ ^o	495.6160	495.4777	604.8886	4.378	2.900e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).